



Dr. N.G.P. Institute of Technology, Coimbatore – 641048

(An Autonomous Institution)

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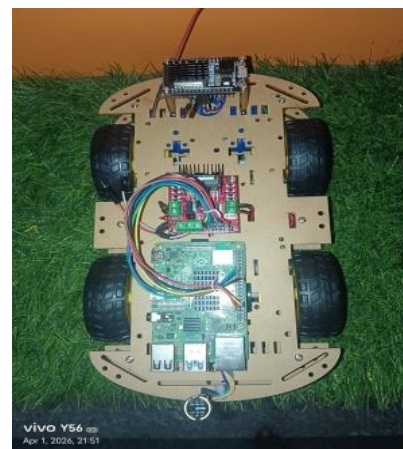
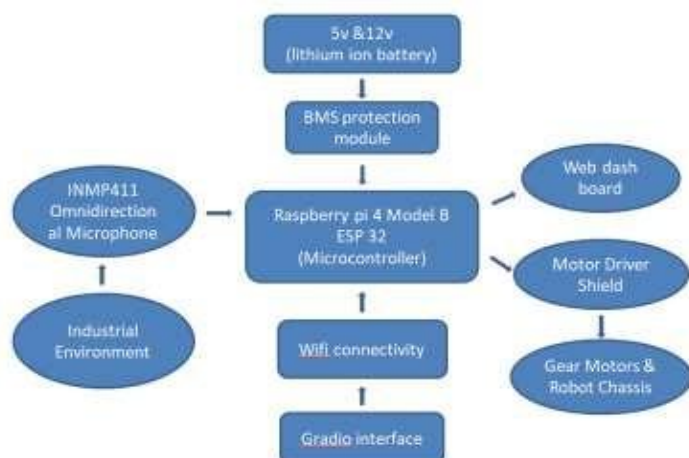
Recognized by UGC & Accredited by NAAC with A+ Grade and

NBA (BME, CSE, ECE, EEE, IT and Mechanical)



**DEPARTMENT OF ELECTRONICS AND COMMUNICATION
ENGINEERING**

**ACOUSTIC BASED MACHINE FAULT DETECTION IN
INDUSTRIES**



DESCRIPTION OF THE PROJECT: The Sonic Shield system is designed to detect abnormal sounds produced by industrial machines and provide early fault detection. Machines such as pumps, fans, sliders, and valves generate specific sound patterns during normal operation, and any mechanical fault causes changes in these patterns. In this project, an omnidirectional microphone is used to capture sound signals from the industrial environment. The captured audio signals are processed and converted into acoustic features for analysis. Machine learning algorithms are applied to classify the sounds as normal or abnormal machine sounds. The sound processing and analysis are performed using the Raspberry Pi 4 Model B, which enables real-time monitoring. The robotic movement of the monitoring system is controlled using the ESP32, allowing the robot to patrol different areas of the industrial environment. When abnormal sounds are detected, alerts are sent through Wi-Fi to a monitoring interface. This system helps industries detect machine faults early, reduce maintenance costs, and improve safety and operational efficiency.

REQUIREMENTS: Raspberry Pi 4 Model B, ESP32 Microcontroller, INMP411 Omnidirectional Microphone, Motor Driver Module, Gear Motors, Robot Chassis, Lithium-ion Battery, Battery Management System (BMS), Micro SD Card.



POs: PO8, PO9, PO10, PO11

PSO: PSO1

PROJECT GUIDE:

STUDENTS MEMBERS: BATCH 2024-2028

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Outcome of the project: Nil